AgriPulse Analytics

A Soil-Test Driven Farm & Financial Decision Engine

Abdullah Munir · 2023-GCUF-074974
Muhammad Hamza · 2023-GCUF-074992
Government Graduate College, Satiana Road, Faisalabad
Department of Computer Science · GCUF

The 30-second story

One farmer. Six acres of loam. One decision he cannot undo.

Four numbers he does not have

cost · water · fertiliser bags · return

He already holds the soil test report. Nothing turns it into those numbers until now.

The decision that cannot be undone

Before sowing	During the season	At harvest
Money is spent	Water is used	Yield arrives
Seed goes in	Fertiliser is applied	Cost is finally known
The choice is locked	The dose is locked	Too late to change

Three problems, all measured in rupees the farmer loses.

Two problems

#	Problem	What it costs him
P1	Decisions made without real information. The lab report is a page of numbers; nobody turns it into a crop choice, fertiliser quantity or cost sheet.	Crop chosen blind. Money and water wasted on the wrong dose.
P2	No planning before harvest. He does not know total expense, water requirement, expected yield or profit.	Learns the cost at season end. Cannot compare two crops.

“Every existing tool asks the farmer for what he already knows.”

Why existing tools do not answer it

MandiOye

25 free calculators incl. fertiliser dose, irrigation water, break-even. But the fertiliser calculator asks you for the nutrient requirement.

Ours takes only the soil reading recommendation bands come from SFRI Guide-V.

SFRI Guide 5(Urdu): Soil & Water Data Interpretation & Fertilizer Recommendations for Farmer Fields (2021)

N-P-K target ranges per soil type; the recommendation method

What we built

React 18 + Vite — 5 pages, 2 chart components
Login/Register · Dashboard · Land & Soil Setup · Planner · NPK Advisor

Express API — 6 route groups, JWT auth, Zod validation
/auth · /ref (districts, crops, soil-types) · /climate · /soil/recommend · /plan/calculate · /simulator/recompute

THE BRAIN — 4 pure calculation services
sfri.service (NPK → bags) · planner.service (cost + water + ROI) · climate.service (suitability)

MongoDB (local)
Compus atlas

User input → published formulas → cited results. 5 pages · 7 screens · 6 routes · 4 services

Screen walkthrough: land, climate, soil

Land & Soil Setup

- District: Faisalabad
- Area: 6.2 acres (0.1–100)
- Soil type: Loam
- Climate : 21°C · 22 mm rain · 34% rain probability · 48% RH · frost risk none
- Soil test: N 42 · P 18 · K 55 kg/ha · pH 7.4

NPK Advisor (radar)

- Axes: pH, K, P, N — ring = your soil, filled = loam target
- N LOW −18 kg/ha
- P LOW −12 kg/ha
- K HIGH — reduce dose
- pH OK
- Source label: SFRI Punjab Guide-V

“K high, reduce dose” — a system that can tell a farmer not to buy something is not selling him anything.

Every number carries its citation

Published: copied from a named public source — open it, find the number. **Calculated:** a published method on a published figure — read the method, check the arithmetic.

#	Source	What we take from it
1	SFRI Punjab — Guide-V	N-P-K target ranges per soil type; the recommendation method
2	PCPA Cotton Book	Per-acre cost breakdown; 2.35 / 1.04 bag validation figure
3	PAU Package of Practices	Seed rates, varieties, sowing windows, district yield bands
4	FAO Irrigation & Drainage Paper 56	Crop coefficients (Kc), ETc method, effective rainfall
5	NFDC Fertiliser Statistics	Retail prices of urea, DAP, MOP
6	World Bank Climate Portal / NASA POWER	Monthly temperature, rainfall, ETo normals 1991–2020
7	Punjab & Sindh Agriculture Depts	District list, crop calendars, soil type distribution
8	Punjab soil testing laboratories	The soil test report the farmer already holds

Tech stack, and the reason for each choice

Layer	Tools (19)	Why
Frontend	React 18, Vite, React Router, Tailwind CSS, Recharts, Fetch API	Fast start; built-in Radar chart; browser-native fetch
Backend	Node.js, Express.js	Pure MERN, one language
Data	MongoDB (local), Compass	Local mongod = offline by construction
Auth & safety	JWT, Bcrypt, Zod, Cors, Dotenv, Morgan, Nodemon	No plain-text passwords; per-user isolation; every input validated
Tooling	Git + GitHub	Version control and review

Entirely local: local DB, local server. No account, no subscription, no hosted service.

What this is, honestly: a proof of concept

Term	Meaning	Is that AgriPulse?
Prototype	Throwaway artefact to explore if an idea is worth pursuing	No pure functions under unit test; production discipline
Proof of concept	Answers one question: can this approach work? Success is a test	Yes can published methods be implemented correctly, locally, and verified against an official figure? Week 5 answers it
Product	Usable by real users, supported, complete	Not yet and we do not claim it

Any Question?